PROBLEM 36

Pendulum String Dragged by a Rotating Bar

Original statement · Problems.pdf, p. 24

Rotating Frame + String Constraint

A heavy point particle P with mass m hangs from a fixed point A on the ceiling with a massless string of constant length L. The particle is initially at rest. A bar b rotates about its center O in a horizontal plane at a distance h from the ceiling, with a known, constant angular velocity ω. The horizontal distance between O and A is R. At time $t = 0$, the bar collides with the string and drags it along its motion. The string slides over the bar without friction. We call C the instantaneous point of contact of the string with the bar. We define a reference frame S_1 that rotates with the bar as shown in the figure. We call ξ the distance between O and C; ℓ the length of the string segment between C and P; θ the angle between the vertical plane Ox_1z_1 and the plane defined by the bar and P; and φ the angle formed between the string segment CP and the bar. You are asked to study only the part of the motion in which the string remains tense and in contact with the bar.

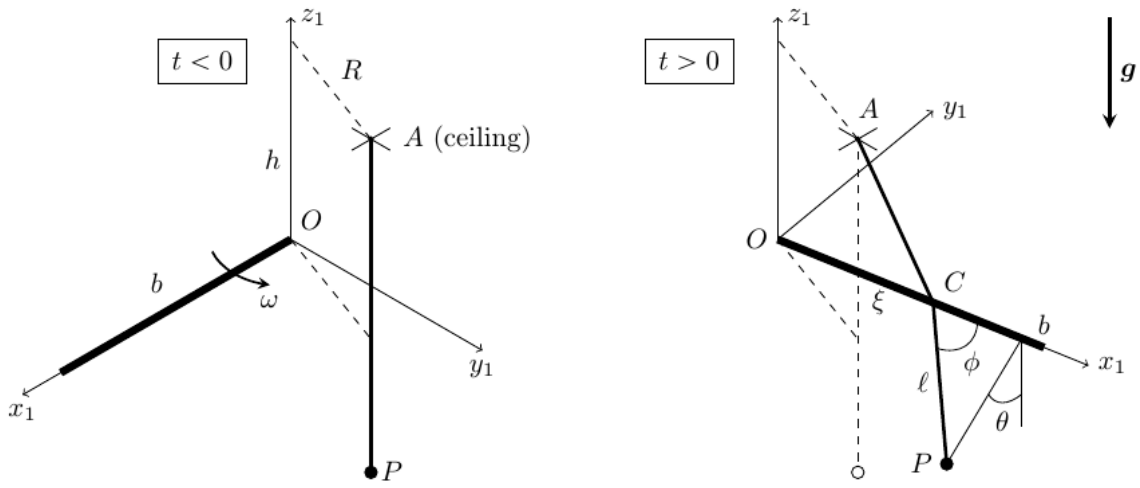


Figure: Left, before the impact (t negative): bar b rotating with ω about O in the horizontal plane x1y1, ceiling point A at height h above that plane and horizontal distance R from O, P hanging below A. Right, after the impact (t positive): frame S1 = Ox1y1z1 attached to the bar, contact point C at distance ξ from O, string A–C–P with segment CP of length ℓ at angle φ from the bar, θ the angle of the plane bar–P from the vertical; gravity g downward (Problems.pdf, p. 24).

(a) Write down the position and velocity of point A in S_1 as a function of time.

(b)

Obtain a general expression for the position, velocity, and acceleration of P in S_1 as a function of $\xi, \ell, \theta, \varphi$, their derivatives, and time.

(c) Express AC, ℓ, and φ as a function of ξ and time.

(d) Identify and express all the forces acting on P.

(e) Write down a set of second order differential equations for the evolution of the unknowns of the problem.

(f) Compute the initial conditions for the previous equations at $t = 0$.

(g) Is the mechanical energy of the particle conserved in time? Justify your answer.

(h) If/when the string separates from the bar, express the geometric condition as a relation between θ and time.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** a heavy particle P (mass m) hangs from the fixed ceiling point A by a string of length L. A bar turning at constant rate ω about the vertical axis through its centre O, in a horizontal plane a distance h below the ceiling, hits the string at $t = 0$ and drags it. The string then runs $A \to C \to P$, bent over the bar at the contact point C.
- **Why the frame S_1 attached to the bar.** In S_1 the bar is the fixed axis Ox_1, so the contact point is simply $C = (\xi, 0, 0)$ and the string geometry is described by $\xi, \ell, \varphi, \theta$. The price is that A moves in S_1 and that the centrifugal and Coriolis forces (Slides 27–28) have to be added. Since $\boldsymbol{\omega}_{10}$ is constant and O is fixed, the Euler force and the force of the origin's acceleration vanish, and the centrifugal force has a potential (Slide 30).

- **Hypotheses:**
  1. A frame S_0 with origin O and vertical axis $\mathbf{k}_0$ (upward) is inertial; gravity is $-g\,\mathbf{k}_0$.
  2. The string is massless and inextensible, and stays tense ($T > 0$) and in contact with the bar (the only phase studied, as the statement asks).
  3. The bar is thin (a straight line) and long enough for C to stay on it; the contact is frictionless.
  4. The rotation of the bar is prescribed: ω is kept constant whatever the string does.
  5. The bar reaches the string at $t = 0$ while P is at rest below A, so $L > h$.
- **Degrees of freedom:** P moves in space with the single constraint "string tense", so it has **two** degrees of freedom. We take ξ and θ; ℓ and φ are fixed by ξ and t through the constraints of part (c). The tension T is the additional unknown.

Symbol	Meaning	SI unit
m	mass of P	kg
L	total length of the string	m
h	height of the ceiling above the plane of the bar	m
R	horizontal distance between O and A	m
ω	angular velocity of the bar (constant)	rad/s
g	gravitational acceleration	m/s^2
ξ	distance OC along the bar	m
ℓ	length of the segment CP	m
φ	angle between $\overrightarrow{CP}$ and the bar ($\mathbf{i}_1$)	rad
θ	angle between the plane Ox_1z_1 and the plane bar–P	rad
T	tension of the string	N

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, $\mathbf{k}_0$ vertical upward and $\mathbf{i}_0$ pointing horizontally from O towards the vertical of A, so that $\mathbf{r}_0^A = R\,\mathbf{i}_0 + h\,\mathbf{k}_0$. $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the bar, $\mathbf{i}_1$ along the bar, $\mathbf{k}_1 = \mathbf{k}_0$. At $t = 0$ the bar is below A, so S_1 coincides with S_0 at that instant.
- **Sense of rotation.** In the left drawing ($t < 0$) the vertical projection of A lies at positive x_1 and positive y_1, so the bar reaches it turning from $\mathbf{i}_1$ towards $\mathbf{j}_1$: $\boldsymbol{\omega}_{10} = +\omega\,\mathbf{k}_1$. (The same sign is implied by the Coriolis force of the official key.) The angle turned by the bar is ωt.
- **Change of basis** (rotation of angle ωt about $\mathbf{k}_0$; columns are $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$$

The columns are unit vectors and pairwise orthogonal ($-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

- **Motion of S_1: $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$** (the centre of the bar is fixed), $\boldsymbol{\omega}_{10} = \omega\,\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$ (ω constant and $\mathbf{k}_1 = \mathbf{k}_0$ fixed).
- **String basis $\mathcal{B}_2 = \{\mathbf{e}_{CP}, \mathbf{e}_\theta, \mathbf{e}_\varphi\}$.** The segment CP makes the angle φ with $\mathbf{i}_1$, and its component perpendicular to the bar lies in the plane bar–P, rotated by θ from the downward vertical; $\theta > 0$ puts P on the side $y_1 < 0$, the side of A for $0 < \omega t < \pi$ (this is the sign convention of the official key):

$$\mathbf{e}_{CP} = \cos\varphi\,\mathbf{i}_1 - \sin\varphi\sin\theta\,\mathbf{j}_1 - \sin\varphi\cos\theta\,\mathbf{k}_1$$

$$\mathbf{e}_\varphi = \frac{\partial \mathbf{e}_{CP}}{\partial\varphi} = -\sin\varphi\,\mathbf{i}_1 - \cos\varphi\sin\theta\,\mathbf{j}_1 - \cos\varphi\cos\theta\,\mathbf{k}_1, \qquad \mathbf{e}_\theta = \frac{1}{\sin\varphi}\frac{\partial \mathbf{e}_{CP}}{\partial\theta} = -\cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1$$

The matrix with columns $\mathbf{e}_{CP}, \mathbf{e}_\theta, \mathbf{e}_\varphi$ in $\mathcal{B}_1$ is

$$[{}_1R_2] = \begin{pmatrix} \cos\varphi & 0 & -\sin\varphi \\ -\sin\varphi\sin\theta & -\cos\theta & -\cos\varphi\sin\theta \\ -\sin\varphi\cos\theta & \sin\theta & -\cos\varphi\cos\theta \end{pmatrix}$$

Expanding along the first row: $\det[{}_1R_2] = \cos\varphi\left(\cos\varphi\cos^2\theta + \cos\varphi\sin^2\theta\right) - \sin\varphi\left(-\sin\varphi\sin^2\theta - \sin\varphi\cos^2\theta\right) = \cos^2\varphi + \sin^2\varphi = 1$. Each column has unit norm, and $\mathbf{e}_{CP}\cdot\mathbf{e}_\theta = \sin\varphi\sin\theta\cos\theta - \sin\varphi\cos\theta\sin\theta = 0$, $\mathbf{e}_{CP}\cdot\mathbf{e}_\varphi = -\cos\varphi\sin\varphi + \sin\varphi\cos\varphi\left(\sin^2\theta + \cos^2\theta\right) = 0$, $\mathbf{e}_\theta\cdot\mathbf{e}_\varphi = \cos\theta\cos\varphi\sin\theta - \sin\theta\cos\varphi\cos\theta = 0$, so $[{}_1R_2][{}_1R_2]^T = I$ and $\mathbf{e}_{CP}\times\mathbf{e}_\theta = \mathbf{e}_\varphi$.

- **Derivatives of $\mathcal{B}_2$ in S_1** (chain rule, $\frac{d}{dt}\mathbf{e}(\varphi(t),\theta(t)) = \frac{\partial\mathbf{e}}{\partial\varphi}\dot\varphi + \frac{\partial\mathbf{e}}{\partial\theta}\dot\theta$, with $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ constant in S_1). Using $\partial\mathbf{e}_\varphi/\partial\varphi = \cos\varphi\,(-\mathbf{i}_1) + \sin\varphi\sin\theta\,\mathbf{j}_1 + \sin\varphi\cos\theta\,\mathbf{k}_1 = -\mathbf{e}_{CP}$, $\partial\mathbf{e}_\varphi/\partial\theta = -\cos\varphi\cos\theta\,\mathbf{j}_1 + \cos\varphi\sin\theta\,\mathbf{k}_1 = \cos\varphi\,\mathbf{e}_\theta$ and $\partial\mathbf{e}_\theta/\partial\theta = \sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1 = -\left(\sin\varphi\,\mathbf{e}_{CP} + \cos\varphi\,\mathbf{e}_\varphi\right)$:

$$\left.\frac{d\mathbf{e}_{CP}}{dt}\right|_1 = \dot\varphi\,\mathbf{e}_\varphi + \dot\theta\sin\varphi\,\mathbf{e}_\theta, \qquad \left.\frac{d\mathbf{e}_\varphi}{dt}\right|_1 = -\dot\varphi\,\mathbf{e}_{CP} + \dot\theta\cos\varphi\,\mathbf{e}_\theta, \qquad \left.\frac{d\mathbf{e}_\theta}{dt}\right|_1 = -\dot\theta\,(\sin\varphi\,\mathbf{e}_{CP} + \cos\varphi\,\mathbf{e}_\varphi)$$

PHASE 3

Step-by-Step Mathematical Deduction

(a) Position and velocity of A in S_1

Pedagogical Justification: A is fixed in S_0; its coordinates in S_1 follow from $\mathbf{r}_1^A = [{}_0R_1]^T\mathbf{r}_0^A$ (the origins coincide), and its relative velocity is the derivative of those coordinates in S_1.

$$[{}_0R_1]^T\begin{pmatrix}R\\0\\h\end{pmatrix} = \begin{pmatrix}R\cos\omega t\\-R\sin\omega t\\h\end{pmatrix} \quad\Longrightarrow\quad \boxed{\mathbf{r}_1^A = R\cos\omega t\,\mathbf{i}_1 - R\sin\omega t\,\mathbf{j}_1 + h\,\mathbf{k}_1}$$

With the chain rule $\frac{d}{dt}\cos\omega t = -\omega\sin\omega t$ and $\frac{d}{dt}\sin\omega t = \omega\cos\omega t$:

$$\boxed{\mathbf{v}_1^A = \left.\frac{d\mathbf{r}_1^A}{dt}\right|_1 = -R\omega\sin\omega t\,\mathbf{i}_1 - R\omega\cos\omega t\,\mathbf{j}_1}$$

Check with the velocity composition (Notes Eq. 3.26): $\mathbf{0} = \mathbf{v}_0^A = \mathbf{v}_1^A + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^A$, and $\omega\mathbf{k}_1\times\mathbf{r}_1^A = R\omega\cos\omega t\,\mathbf{j}_1 + R\omega\sin\omega t\,\mathbf{i}_1 = -\mathbf{v}_1^A$. In S_1, A describes a circle of radius R at height h with angular velocity $-\omega$.

(b) Position, velocity and acceleration of P in S_1

Pedagogical Justification: $\overrightarrow{OP} = \overrightarrow{OC} + \overrightarrow{CP}$ with $\overrightarrow{OC} = \xi\,\mathbf{i}_1$ along the bar and $\overrightarrow{CP} = \ell\,\mathbf{e}_{CP}$. Derivatives are taken in S_1 with the product rule and the basis derivatives of Phase 2.

Position:

$$\boxed{\mathbf{r}_1^P = \xi\,\mathbf{i}_1 + \ell\,\mathbf{e}_{CP} = (\xi + \ell\cos\varphi)\,\mathbf{i}_1 - \ell\sin\varphi\sin\theta\,\mathbf{j}_1 - \ell\sin\varphi\cos\theta\,\mathbf{k}_1}$$

Velocity. Product rule, $\frac{d}{dt}(\ell\,\mathbf{e}_{CP}) = \dot\ell\,\mathbf{e}_{CP} + \ell\left(\dot\varphi\,\mathbf{e}_\varphi + \dot\theta\sin\varphi\,\mathbf{e}_\theta\right)$:

$$\boxed{\mathbf{v}_1^P = \dot{\xi}\,\mathbf{i}_1 + \dot{\ell}\,\mathbf{e}_{CP} + \ell\dot{\varphi}\,\mathbf{e}_\varphi + \ell\dot{\theta}\sin\varphi\,\mathbf{e}_\theta}$$

In $\mathcal{B}_1$ (differentiating each component; e.g. $\frac{d}{dt}(-\ell\sin\varphi\sin\theta) = -\dot{\ell}\sin\varphi\sin\theta - \ell\dot{\varphi}\cos\varphi\sin\theta - \ell\dot{\theta}\sin\varphi\cos\theta$):

$$\mathbf{v}_1^P = \left(\dot{\xi} + \dot{\ell}\cos\varphi - \ell\dot{\varphi}\sin\varphi\right)\mathbf{i}_1 - \left(\dot{\ell}\sin\varphi\sin\theta + \ell\dot{\varphi}\cos\varphi\sin\theta + \ell\dot{\theta}\sin\varphi\cos\theta\right)\mathbf{j}_1 - \left(\dot{\ell}\sin\varphi\cos\theta + \ell\dot{\varphi}\cos\varphi\cos\theta - \ell\dot{\theta}\sin\varphi\sin\theta\right)\mathbf{k}_1$$

Acceleration. Differentiate term by term:

- $\frac{d}{dt}(\dot{\xi}\,\mathbf{i}_1) = \ddot{\xi}\,\mathbf{i}_1$;
- $\frac{d}{dt}(\dot{\ell}\,\mathbf{e}_{CP}) = \ddot{\ell}\,\mathbf{e}_{CP} + \dot{\ell}\dot{\varphi}\,\mathbf{e}_\varphi + \dot{\ell}\dot{\theta}\sin\varphi\,\mathbf{e}_\theta$;
- $\frac{d}{dt}(\ell\dot{\varphi}\,\mathbf{e}_\varphi) = (\dot{\ell}\dot{\varphi} + \ell\ddot{\varphi})\,\mathbf{e}_\varphi + \ell\dot{\varphi}\left(-\dot{\varphi}\,\mathbf{e}_{CP} + \dot{\theta}\cos\varphi\,\mathbf{e}_\theta\right)$;
- $\frac{d}{dt}(\ell\dot{\theta}\sin\varphi\,\mathbf{e}_\theta) = \left(\dot{\ell}\dot{\theta}\sin\varphi + \ell\ddot{\theta}\sin\varphi + \ell\dot{\theta}\dot{\varphi}\cos\varphi\right)\mathbf{e}_\theta - \ell\dot{\theta}^2\sin\varphi\,(\sin\varphi\,\mathbf{e}_{CP} + \cos\varphi\,\mathbf{e}_\varphi)$.

Collecting along each unit vector:

$$\boxed{\mathbf{a}_1^P = \ddot{\xi}\,\mathbf{i}_1 + \left(\ddot{\ell} - \ell\dot{\varphi}^2 - \ell\dot{\theta}^2\sin^2\varphi\right)\mathbf{e}_{CP} + \left(\ell\ddot{\varphi} + 2\dot{\ell}\dot{\varphi} - \ell\dot{\theta}^2\sin\varphi\cos\varphi\right)\mathbf{e}_\varphi + \left(\ell\ddot{\theta}\sin\varphi + 2\dot{\ell}\dot{\theta}\sin\varphi + 2\ell\dot{\theta}\dot{\varphi}\cos\varphi\right)\mathbf{e}_\theta}$$

with $\mathbf{e}_{CP}, \mathbf{e}_\varphi, \mathbf{e}_\theta$ given in $\mathcal{B}_1$ in Phase 2. The bracketed terms are the spherical-coordinate acceleration of P about the moving centre C, plus the acceleration $\ddot{\xi}\,\mathbf{i}_1$ of the centre along the bar. Time enters through ξ, ℓ, φ, θ only; the explicit time dependence appears when ℓ and φ are replaced by the constraints of part (c).

(c) AC, ℓ and φ as functions of ξ and t

> **Pedagogical Justification:** The string has constant length, which gives ℓ. The angle φ follows from the frictionless contact: the massless piece of string at C must be in equilibrium under the two tensions and the bar's reaction, which has no component along the bar.

Vector $\overrightarrow{AC}$: with $\mathbf{r}_1^C = \xi\,\mathbf{i}_1$ and part (a),

$$\boxed{\overrightarrow{AC} = \mathbf{r}_1^C - \mathbf{r}_1^A = (\xi - R\cos\omega t)\,\mathbf{i}_1 + R\sin\omega t\,\mathbf{j}_1 - h\,\mathbf{k}_1}$$

$$AC^2 = (\xi - R\cos\omega t)^2 + R^2\sin^2\omega t + h^2 = \xi^2 - 2R\xi\cos\omega t + R^2\left(\cos^2\omega t + \sin^2\omega t\right) + h^2 \quad\Longrightarrow\quad \boxed{AC = \sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}}$$

Length ℓ. $AC + CP = L$:

$$\boxed{\ell = L - \sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}}$$

Angle φ. With no friction the tension is the same, T, on both sides of C. Equilibrium of the massless string element at C: $T\,\mathbf{e}_{CA} + T\,\mathbf{e}_{CP} + \mathbf{N}_b = \mathbf{0}$ with $\mathbf{N}_b \cdot \mathbf{i}_1 = 0$. Projecting on $\mathbf{i}_1$: $\mathbf{e}_{CP}\cdot\mathbf{i}_1 = -\mathbf{e}_{CA}\cdot\mathbf{i}_1$, where $\mathbf{e}_{CA} = -\overrightarrow{AC}/AC$. Hence

$$\cos\varphi = \frac{\xi - R\cos\omega t}{AC} \quad\Longrightarrow\quad \boxed{\varphi = \arccos\frac{\xi - R\cos\omega t}{\sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}}}, \qquad \sin\varphi = \frac{\sqrt{R^2\sin^2\omega t + h^2}}{AC}$$

($\varphi \in (0, \pi)$, so $\sin\varphi > 0$). The two segments make equal angles with the bar, on opposite sides of the plane perpendicular to it. (Extension: this is also the condition for the path $A \to C \to P$ to be the shortest one for fixed P, $\frac{\partial}{\partial\xi}(AC + CP) = \cos\angle(\overrightarrow{AC}, \mathbf{i}_1) - \cos\varphi = 0$.)

Compact notation for part (e). Let $q = \xi - R\cos\omega t$, $\rho = \sqrt{R^2\sin^2\omega t + h^2}$ and $D = AC = \sqrt{q^2 + \rho^2}$. Then $q = D\cos\varphi$, $\rho = D\sin\varphi$: (D, φ) are the polar coordinates of the point (q, ρ), and $\ell = L - D$. Their derivatives:

- $\dot q = \dot\xi + R\omega\sin\omega t$ and $\ddot q = \ddot\xi + R\omega^2\cos\omega t$ (chain rule on $\cos\omega t$).
- From $\rho^2 = R^2\sin^2\omega t + h^2$: $2\rho\dot\rho = 2R^2\omega\sin\omega t\cos\omega t$, so $\dot\rho = \dfrac{R^2\omega\sin 2\omega t}{2\rho}$.
  Differentiating $\rho\dot\rho = \frac{1}{2}R^2\omega\sin 2\omega t$ (product rule): $\dot\rho^2 + \rho\ddot\rho = R^2\omega^2\cos 2\omega t$, so $\ddot\rho = \dfrac{R^2\omega^2\cos 2\omega t - \dot\rho^2}{\rho}$.
- Differentiating $q = D\cos\varphi$, $\rho = D\sin\varphi$: $\dot q = \dot D\cos\varphi - D\dot\varphi\sin\varphi$, $\dot\rho = \dot D\sin\varphi + D\dot\varphi\cos\varphi$. Solving this rotation-type system:

$$\dot D = \dot q\cos\varphi + \dot\rho\sin\varphi, \qquad D\dot\varphi = \dot\rho\cos\varphi - \dot q\sin\varphi$$

- Differentiating once more, $\ddot q = (\ddot D - D\dot\varphi^2)\cos\varphi - (D\ddot\varphi + 2\dot D\dot\varphi)\sin\varphi$ and $\ddot\rho = (\ddot D - D\dot\varphi^2)\sin\varphi + (D\ddot\varphi + 2\dot D\dot\varphi)\cos\varphi$, so

$$\ddot{D} - D\dot{\varphi}^2 = \ddot{q}\cos\varphi + \ddot{\rho}\sin\varphi, \qquad D\ddot{\varphi} + 2\dot{D}\dot{\varphi} = \ddot{\rho}\cos\varphi - \ddot{q}\sin\varphi$$

- Finally $\dot{\ell} = -\dot{D}, \ddot{\ell} = -\ddot{D}$. Since $\dot{\rho}\sin\varphi = \dfrac{R^2\omega\sin\omega t\cos\omega t}{\rho}\,\dfrac{\rho}{D}$ and $\cos\varphi = q/D$,

$$\dot{\ell} = -(\dot{\xi} + R\omega\sin\omega t)\frac{\xi - R\cos\omega t}{D} - \frac{R^2\omega\sin\omega t\cos\omega t}{D} \quad\Longrightarrow\quad \boxed{\dot{\ell} + \dot{\xi}\cos\varphi = -\frac{R\omega\,\xi\sin\omega t}{D}}$$

(d) Forces acting on P

> **Pedagogical Justification:** In the non-inertial frame S_1, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63). The actual forces on P are only the weight and the pull of the string (the bar touches the string, not P). The inertia forces are those of Slide 28 with $\mathbf{a}_0^O = \mathbf{0}, \boldsymbol{\alpha}_{10} = \mathbf{0}, \boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$.

Actual forces:

- Weight: $\mathbf{W} = -mg\,\mathbf{k}_1$.
- Tension, along the string towards C: $\mathbf{T} = -T\,\mathbf{e}_{CP} = -T\cos\varphi\,\mathbf{i}_1 + T\sin\varphi\sin\theta\,\mathbf{j}_1 + T\sin\varphi\cos\theta\,\mathbf{k}_1$, with $T > 0$.

Inertia forces: write $\mathbf{r}_1^P = x\,\mathbf{i}_1 + y\,\mathbf{j}_1 + z\,\mathbf{k}_1$ with $x = \xi + \ell\cos\varphi, y = -\ell\sin\varphi\sin\theta$, and $\mathbf{v}_1^P = v_x\mathbf{i}_1 + v_y\mathbf{j}_1 + v_z\mathbf{k}_1$ from part (b). With $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1, \mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1, \mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$:

- Origin and Euler: $\mathbf{F}_A = -m\mathbf{a}_0^O = \mathbf{0}, \mathbf{F}_E = -m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P = \mathbf{0}$.
- Centrifugal: $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \omega\,(x\,\mathbf{j}_1 - y\,\mathbf{i}_1), \boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = \omega^2\,(-x\,\mathbf{i}_1 - y\,\mathbf{j}_1)$, so (Slide 30, $\mathbf{r}_\perp = x\,\mathbf{i}_1 + y\,\mathbf{j}_1$)

$$\boxed{\mathbf{F}_C = m\omega^2(\xi + \ell\cos\varphi)\,\mathbf{i}_1 - m\omega^2\ell\sin\varphi\sin\theta\,\mathbf{j}_1}$$

- Coriolis: $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = \omega\,(v_x\,\mathbf{j}_1 - v_y\,\mathbf{i}_1)$, so $\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = 2m\omega v_y\,\mathbf{i}_1 - 2m\omega v_x\,\mathbf{j}_1$:

$$\boxed{\mathbf{F}_{COR} = -2m\omega\left(\dot{\ell}\sin\varphi\sin\theta + \ell\dot{\varphi}\cos\varphi\sin\theta + \ell\dot{\theta}\sin\varphi\cos\theta\right)\mathbf{i}_1 - 2m\omega\left(\dot{\xi} + \dot{\ell}\cos\varphi - \ell\dot{\varphi}\sin\varphi\right)\mathbf{j}_1}$$

Both inertia forces agree with the official key.

(e) Equations of motion

> **Pedagogical Justification:** Project $m\mathbf{a}_1^P = \mathbf{W} + \mathbf{T} + \mathbf{F}_C + \mathbf{F}_{COR}$ on $\mathcal{B}_2$. The tension is along $\mathbf{e}_{CP}$, so the projections on $\mathbf{e}_\theta$ and $\mathbf{e}_\varphi$ are free of the unknown T: they are the two equations of motion for the two degrees of freedom ξ, θ. The projection on $\mathbf{e}_{CP}$ then gives T .

Projections of each term (using $\mathbf{i}_1 \cdot \mathbf{e}_{CP} = \cos\varphi$, $\mathbf{i}_1 \cdot \mathbf{e}_\varphi = -\sin\varphi$, $\mathbf{i}_1 \cdot \mathbf{e}_\theta = 0$, and the components of Phase 2):

Term	$\cdot\,\mathbf{e}_\theta$	$\cdot\,\mathbf{e}_\varphi$	$\cdot\,\mathbf{e}_{CP}$
$\mathbf{a}_1^P$	$\ell\ddot{\theta}\sin\varphi + 2\dot{\ell}\dot{\theta}\sin\varphi + 2\ell\dot{\theta}\dot{\varphi}\cos\varphi$	$\ell\ddot{\varphi} + 2\dot{\ell}\dot{\varphi} - \ddot{\xi}\sin\varphi - \ell\dot{\theta}^2\sin\varphi\cos\varphi$	$\ddot{\xi}\cos\varphi + \ddot{\ell} - \ell\dot{\varphi}^2 - \ell\dot{\theta}^2\sin^2\varphi$
$\mathbf{W}$	$-mg\sin\theta$	$mg\cos\varphi\cos\theta$	$mg\sin\varphi\cos\theta$
$\mathbf{T}$	0	0	$-T$
$\mathbf{F}_C$	$m\omega^2\ell\sin\varphi\sin\theta\cos\theta$	$-m\omega^2\sin\varphi\left(\xi + \ell\cos\varphi\cos^2\theta\right)$	$m\omega^2\left[(\xi + \ell\cos\varphi)\cos\varphi + \ell\sin^2\varphi\sin^2\theta\right]$
$\mathbf{F}_{COR}$	$2m\omega\cos\theta\, v_x$	$2m\omega\left[\sin\theta(\dot{\ell} + \dot{\xi}\cos\varphi) + \ell\dot{\theta}\sin^2\varphi\cos\theta\right]$	$2m\omega\left[\sin\theta(\dot{\xi}\sin\varphi - \ell\dot{\varphi}) - \ell\dot{\theta}\sin\varphi\cos\varphi\cos\theta\right]$

Details of the less obvious entries:

- $\mathbf{F}_C \cdot \mathbf{e}_\varphi = m\omega^2\left[-x\sin\varphi - y\cos\varphi\sin\theta\right] = m\omega^2\left[-(\xi + \ell\cos\varphi)\sin\varphi + \ell\sin\varphi\cos\varphi\sin^2\theta\right] = -m\omega^2\sin\varphi\left[\xi + \ell\cos\varphi(1 - \sin^2\theta)\right]$.
- $\mathbf{F}_{COR} \cdot \mathbf{e}_\varphi = 2m\omega\left[-v_y\sin\varphi + v_x\cos\varphi\sin\theta\right]$, where $-v_y\sin\varphi = \dot{\ell}\sin^2\varphi\sin\theta + \ell\dot{\varphi}\sin\varphi\cos\varphi\sin\theta + \ell\dot{\theta}\sin^2\varphi\cos\theta$ and $v_x\cos\varphi\sin\theta = \dot{\xi}\cos\varphi\sin\theta + \dot{\ell}\cos^2\varphi\sin\theta - \ell\dot{\varphi}\sin\varphi\cos\varphi\sin\theta$; the $\dot{\varphi}$ terms cancel and $\sin^2\varphi + \cos^2\varphi = 1$.
- $\mathbf{F}_{COR} \cdot \mathbf{e}_{CP} = 2m\omega\left[v_y\cos\varphi + v_x\sin\varphi\sin\theta\right]$: the $\dot{\ell}$ terms cancel and the $\dot{\varphi}$ terms add up to $-\ell\dot{\varphi}\sin\theta\left(\cos^2\varphi + \sin^2\varphi\right)$.
- $\mathbf{F}_{COR} \cdot \mathbf{e}_\theta = 2m\omega\left[-v_x\left(-\cos\theta\right)\right] = 2m\omega\cos\theta\, v_x$, with $v_x = \dot{\xi} + \dot{\ell}\cos\varphi - \ell\dot{\varphi}\sin\varphi$.

Equation along $\mathbf{e}_\theta$ (divided by m):

$$\boxed{\ell\sin\varphi\,\ddot{\theta} + 2\dot{\ell}\dot{\theta}\sin\varphi + 2\ell\dot{\theta}\dot{\varphi}\cos\varphi = -g\sin\theta + \omega^2\ell\sin\varphi\sin\theta\cos\theta + 2\omega\cos\theta\left(\dot{\xi} + \dot{\ell}\cos\varphi - \ell\dot{\varphi}\sin\varphi\right)}$$

Equation along $\mathbf{e}_\varphi$ (divided by m):

$$\boxed{\ell\ddot{\varphi} + 2\dot{\ell}\dot{\varphi} - \ddot{\xi}\sin\varphi - \ell\dot{\theta}^2\sin\varphi\cos\varphi = g\cos\varphi\cos\theta - \omega^2\sin\varphi\left(\xi + \ell\cos\varphi\cos^2\theta\right) + 2\omega\left[\sin\theta\left(\dot{\ell} + \dot{\xi}\cos\varphi\right) + \ell\dot{\theta}\sin^2\varphi\cos\theta\right]}$$

Closing the system. ℓ, φ and their derivatives are the functions of $(\xi, \dot\xi, \ddot\xi, t)$ of part (c). In particular $\ell\ddot\varphi = \frac{\ell}{D}\left(\ddot\rho\cos\varphi - \ddot q\sin\varphi - 2\dot D\dot\varphi\right)$ with $\ddot q = \ddot\xi + R\omega^2\cos\omega t$, and $2\dot\ell\dot\varphi = -2\dot D\dot\varphi$, so with $\ell + D = L$ the left-hand side of the $\mathbf{e}_\varphi$ equation becomes

$$\ell\ddot\varphi + 2\dot\ell\dot\varphi - \ddot\xi\sin\varphi = -\frac{L}{D}\sin\varphi\,\ddot\xi + \frac{\ell}{D}\left(\ddot\rho\cos\varphi - R\omega^2\cos\omega t\sin\varphi\right) - \frac{2L}{D}\dot D\dot\varphi$$

The coefficient of $\ddot\xi$, $-(L/D)\sin\varphi$, is non-zero while the string is bent ($\sin\varphi > 0$), so the $\mathbf{e}_\varphi$ equation gives $\ddot\xi$ and the $\mathbf{e}_\theta$ equation gives $\ddot\theta$: **two second-order ODEs for $\xi(t)$ and $\theta(t)$**, non-autonomous through ωt (the motion of A in S_1). In the $\mathbf{e}_\varphi$ equation the Coriolis term can also be written with $\dot\ell + \dot\xi\cos\varphi = -R\omega\xi\sin\omega t/D$.

Tension (projection on $\mathbf{e}_{CP}$, solved for T):

$$\boxed{T = -m\left(\ddot\xi\cos\varphi + \ddot l - \ell\dot\varphi^2 - \ell\dot\theta^2\sin^2\varphi\right) + mg\sin\varphi\cos\theta + m\omega^2\left[(\xi + \ell\cos\varphi)\cos\varphi + \ell\sin^2\varphi\sin^2\theta\right] + 2m\omega\left[\sin\theta\left(\dot\xi\sin\varphi - \ell\dot\varphi\right) - \ell\dot\theta\sin\varphi\cos\varphi\cos\theta\right]}$$

The solution is valid while $T > 0$ (string tense).

(f) Initial conditions at $t = 0$

> **Pedagogical Justification:** At $t = 0^-$ the particle hangs at rest below A, and the string is straight and vertical. The bar touches it at the point of the string at the height of the bar. No impulsive force acts on P: the string is perpendicular to the bar, so displacing C sideways does not change the path length to first order ($\dot D(0) = 0$ below). Position and velocity of P are therefore continuous at $t = 0$.

Positions. At $t = 0$ the bar lies along $\mathbf{i}_0$, and the string meets it directly below A: $\xi(0) = R$. Then $q(0) = R - R = 0$, $\rho(0) = h$, $D(0) = h$, so

$$\xi(0) = R, \qquad \ell(0) = L - h, \qquad \cos\varphi(0) = 0 \;\Rightarrow\; \varphi(0) = \frac{\pi}{2}, \qquad \theta(0) = 0$$

(P is directly below the bar in the plane Ox_1z_1). Check: $\mathbf{r}_1^P(0) = R\,\mathbf{i}_1 - (L - h)\,\mathbf{k}_1$, i.e. P at depth $L - h$ below the bar, L below A.

Velocities. $\mathbf{v}_0^P(0) = \mathbf{0}$, and by Notes Eq. 3.26

$$\mathbf{v}_1^P(0) = \mathbf{v}_0^P - \boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = -\omega\,\mathbf{k}_1\times\left[R\,\mathbf{i}_1 - (L - h)\,\mathbf{k}_1\right] = -\omega R\,\mathbf{j}_1$$

At $\varphi = \pi/2, \theta = 0$: $\mathbf{e}_{CP} = -\mathbf{k}_1, \mathbf{e}_\varphi = -\mathbf{i}_1, \mathbf{e}_\theta = -\mathbf{j}_1$, so the velocity of part (b) is $\mathbf{v}_1^P(0) = \left(\dot\xi - \ell\dot\varphi\right)\mathbf{i}_1 - \ell\dot\theta\,\mathbf{j}_1 - \dot\ell\,\mathbf{k}_1$. Matching components:

- $\mathbf{k}_1$: $\dot\ell(0) = 0$, consistent with the constraint: $\dot D(0) = \dot q\cos\frac{\pi}{2} + \dot\rho(0)\sin\frac{\pi}{2} = \dot\rho(0) = \dfrac{R^2\omega\sin 0}{2h} = 0$.
- $\mathbf{j}_1$: $(L-h)\,\dot\theta(0) = \omega R$, so $\dot\theta(0) = \dfrac{\omega R}{L-h}$.
- $\mathbf{i}_1$: $\dot\xi(0) = (L-h)\,\dot\varphi(0)$. From part (c), $D\dot\varphi = \dot\rho\cos\varphi - \dot q\sin\varphi$ gives $h\,\dot\varphi(0) = -\dot q(0) = -\left(\dot\xi(0) + R\omega\sin 0\right) = -\dot\xi(0)$. Substituting, $\dot\xi(0)\left(1 + \dfrac{L-h}{h}\right) = \dot\xi(0)\dfrac{L}{h} = 0$, so $\dot\xi(0) = 0$ and $\dot\varphi(0) = 0$.

$$\boxed{\xi(0) = R, \quad \dot\xi(0) = 0, \quad \theta(0) = 0, \quad \dot\theta(0) = \frac{\omega R}{L-h}} \qquad \left(\ell(0) = L-h,\ \dot\ell(0) = 0,\ \varphi(0) = \frac{\pi}{2},\ \dot\varphi(0) = 0\right)$$

In S_1 the particle starts moving backwards ($-\mathbf{j}_1$) with the speed ωR of the point of the bar that hits the string: P lags behind the bar, θ grows from zero.

(g) Is the mechanical energy conserved?

> **Pedagogical Justification:** By Slide 29, E_1^P is conserved only if every force that does work in S_1 derives from a potential that does not depend explicitly on time. Here $\mathbf{W}$ and $\mathbf{F}_C$ have potentials (mgz and $-\frac{1}{2}m\omega^2 r_\perp^2$, ω constant, Slide 30) and $\mathbf{F}_{COR}$ does no work (Slide 31). The question is the power of the tension, which is a constraint force of a constraint that moves.

In S_1. Define $E_1^P = \frac{1}{2}m\left|\mathbf{v}_1^P\right|^2 + mgz - \frac{1}{2}m\omega^2\left(x^2+y^2\right)$. By the work–energy theorem in S_1, $\dfrac{dE_1^P}{dt} = \mathbf{T}\cdot\mathbf{v}_1^P = -T\,\mathbf{e}_{CP}\cdot\mathbf{v}_1^P$. From part (b), $\mathbf{e}_{CP}\cdot\mathbf{v}_1^P = \dot\xi\cos\varphi + \dot\ell$, and part (c) gives $\dot\ell + \dot\xi\cos\varphi = -R\omega\xi\sin\omega t/D$. Therefore

$$\boxed{\frac{dE_1^P}{dt} = \frac{T\,R\,\omega\,\xi\sin\omega t}{\sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}}}$$

which is positive for $0 < \omega t < \pi$ (with $T > 0, \xi > 0$): E_1^P **is not conserved.** The reason is that the end A of the string moves in S_1 (part (a)): the constraint "string tense" depends explicitly on time in S_1, and the tension does work on P.

In S_0. $E_0^P = \frac{1}{2}m\left|\mathbf{v}_0^P\right|^2 + mgz$ changes at the rate $\mathbf{T}\cdot\mathbf{v}_0^P$ with $\mathbf{v}_0^P = \mathbf{v}_1^P + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^P$ (Notes Eq. 3.26). The extra term is $-T\,\mathbf{e}_{CP}\cdot\omega\,(x\,\mathbf{j}_1 - y\,\mathbf{i}_1) = -T\omega\,(-x\sin\varphi\sin\theta - y\cos\varphi) = T\omega\sin\varphi\sin\theta\,(\xi + \ell\cos\varphi - \ell\cos\varphi) = T\omega\xi\sin\varphi\sin\theta$, so

$$\frac{dE_0^P}{dt} = T\,\omega\,\xi\left(\frac{R\sin\omega t}{D} + \sin\varphi\sin\theta\right) \neq 0$$

No: the mechanical energy of the particle is not conserved in either frame. The motor that keeps the bar at constant ω feeds energy into the particle through the string.

(h) Separation condition

> **Pedagogical Justification:** The string can only be pushed by the bar. The bar's force on the string, $\mathbf{N}_b = -T\,(\mathbf{e}_{CA} + \mathbf{e}_{CP})$, vanishes when the string is straight; beyond that instant the bend would have to reverse, which a bar pushing from one side cannot impose. Separation is therefore the instant at which A, C and P become collinear: $\mathbf{e}_{CP} = -\mathbf{e}_{CA}$.

From part (c), $-\mathbf{e}_{CA} = \overrightarrow{AC}/D = \dfrac{1}{D}\left[(\xi - R\cos\omega t)\,\mathbf{i}_1 + R\sin\omega t\,\mathbf{j}_1 - h\,\mathbf{k}_1\right]$. The $\mathbf{i}_1$ components of $\mathbf{e}_{CP}$ and $-\mathbf{e}_{CA}$ are already equal (frictionless condition, $\cos\varphi = q/D$). Equating the other two, with $\sin\varphi = \rho/D$:

$$-\frac{\rho}{D}\sin\theta = \frac{R\sin\omega t}{D}, \qquad -\frac{\rho}{D}\cos\theta = -\frac{h}{D} \quad\Longrightarrow\quad \sin\theta = -\frac{R\sin\omega t}{\rho}, \quad \cos\theta = \frac{h}{\rho}$$

Dividing the two relations ($\cos\theta = h/\rho > 0$):

$$\boxed{\tan\theta = -\frac{R\sin\omega t}{h}} \qquad \text{i.e.} \quad \theta_{sep}(t) = -\arctan\left(\frac{R\sin\omega t}{h}\right)$$

The string leaves the bar at the first instant at which $\theta(t)$, obtained from the equations of part (e), reaches this value. For $0 < \omega t < \pi$ it is negative: P must have swung forward, past the vertical plane of the bar, to the side opposite to A, so that the straight line from A through C passes through P.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Mechanism.** In S_1 the anchor A circles backwards at radius R and speed ωR, while the string is held on the rotating bar. The string's end at A and the bar together act as a moving constraint that pumps energy into the pendulum ($dE_1^P/dt > 0$ for $0 < \omega t < \pi$). The centrifugal force pushes P outward along the bar ($+\mathbf{i}_1$); the Coriolis force deflects it sideways and couples ξ and θ.

- **Initial kick.** $\dot\theta(0) = \omega R/(L-h)$: in S_1 the particle starts swinging backwards about the contact point with the speed of the point of the bar that hits the string, a pendulum of length $L-h$.
- **Limits.** $R \to 0$: A lies on the rotation axis and is fixed in S_1, the constraint becomes time-independent, and $dE_1^P/dt = TR\omega\xi \sin\omega t/D \to 0$: E_1^P becomes a first integral (Slide 29). $\omega \to 0$: the inertia forces vanish and the bar would never reach the string; the problem reduces to a pendulum at rest. Separation: $h \to \infty$ gives $\theta_{sep} \to 0$; $\omega t = \pi$ (anchor on the opposite side of the bar) gives $\theta_{sep} = 0$.
- **Agreement with the key.** The centrifugal and Coriolis forces of part (d) coincide term by term with the official key (the key gives only these two forces).
- **Inertial check.** Differentiating $[{}_0R_1]\mathbf{r}_1^P$ twice reproduces Notes Eq. 3.27; at random states, with $\ddot\xi$, $\ddot\theta$ taken from the two equations of part (e), $m\mathbf{a}_0^P - \mathbf{W}$ is parallel to $\mathbf{e}_{CP}$ (only the tension is left), and $dE_1^P/dt = -T\,\mathbf{e}_{CP}\cdot\mathbf{v}_1^P$; the constraint derivatives of part (c), the initial conditions and the separation condition were also checked symbolically with sympy.
- **Dimensions.** $[AC] = \sqrt{\mathrm{m}^2} = \mathrm{m}$; $[\omega^2\ell] = [\omega\dot\xi] = [g] = \mathrm{m}\cdot\mathrm{s}^{-2}$, the same as $[\ell\ddot\theta] = [\ddot\xi]$ in the equations of part (e); $[m\omega^2\xi] = [m\omega\dot\ell] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$; $[\dot\theta(0)] = \dfrac{\mathrm{s}^{-1}\cdot\mathrm{m}}{\mathrm{m}} = \mathrm{s}^{-1}$; $[TR\omega\xi/D] = \dfrac{\mathrm{N}\cdot\mathrm{m}\cdot\mathrm{s}^{-1}\cdot\mathrm{m}}{\mathrm{m}} = \mathrm{W}$, a power; $[R\sin\omega t/h] = 1$, as $\tan\theta$ must be.

Source: Problems.pdf (PDF page 24); key on PDF page 75